

Vaishnavi Sawant

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EDUCATION

Masters of Technology - Electronic Design and Technology	2023 - 2025
- National Institute of Technology, Calicut	8.84 CGPA
Bachelor of Technology - Electronics and Telecommunication Engineering	2017 – 2021
- Government College of Engineering Karad, Maharashtra	8.60 CGPA

EXPERIENCE

Co/op Intern | AMD, Bengaluru (January – July 2025)

- Integrated multiple IPs for **NBIO**, ensuring functional and architectural correctness.
- Performed CDC, lint, and connectivity checks as part of RTL verification signoff flow, identifying and resolving design issues early in the development cycle.
- Executed regression testing to ensure stability and correctness of RTL designs across multiple test environments.
- Contributed to coverage closure by analyzing toggle coverage and adding functional test scenarios.
- Built automation scripts (Python, Bash, TCL) to
 - Extract tool configurations and shared components from source files, eliminating manual updates.
 - Implement credit alignment and synchronization across multiple IP blocks, enabling seamless flow control and preventing buffer overflows in credit-based protocols.
 - Wrote SystemVerilog assertions (SVA) to validate credit counter behaviour and boundary conditions across IP blocks.

SKILLS

Technical Skills: Digital Design, Verilog, System Verilog, UVM, Assertion, Coverage, Static timing analysis (STA), CDC, Lint, C Programming.

Tools: Xilinx Vivado, Synopsys Verdi.

Subjects: Digital Electronics, VLSI Architecture, Digital IC Design, FPGA, Hardware Software Codesign.

PUBLICATIONS

Hardware-Efficient Matrix Multiplication using Tiled Singular Value Decomposition

Presented paper at VLSI Design and Test (VDAT) 2025 Conference: “Hardware-Efficient Matrix Multiplication using Tiled Singular Value Decomposition” – introduced two novel architectures based on the Hestenes-Jacobi algorithm, achieving up to 90% area and power reduction in Cadence Genus (GPDK 90nm @100 MHz) (Accepted for publication).

PROJECTS

Implementation Of Hardware-Efficient Matrix Multiplication using Tiled Singular Value Decomposition (May – Dec 2024)

- Designed and implemented two novel hardware architectures using Tiled Singular Value Decomposition (TSVD) to accelerate matrix multiplication for deep neural network (DNN) workloads.
- Achieved major hardware optimizations: • Hestenes-Tile Fusion Architecture: ~75% area, 77% power, and 69% gate count reduction. • Hybrid SVD-Tiled Architecture: ~90% reduction in area, power, and gates, with improved latency and throughput.
- Validated and synthesized designs in Cadence Genus (GPDK 90nm, 100 MHz), ensuring functionality and performance.

Design of I2C Communication protocol using Verilog and Verification using System Verilog (May – June 2024)

- Designed and verified an I2C master interface supporting 7-bit addressing, read/write operations, clock stretching, and ACK/NACK handshaking for reliable data transfer.
- Built a constrained-random testbench (Generator/Driver/Monitor/Scoreboard) with mailbox and event synchronization, featuring a self-checking scoreboard to validate read/write data integrity across 20 randomized transactions.

Implementation of Asynchronous and Synchronous FIFO in Verilog and Verification of Synchronous FIFO using System Verilog (March – April 2024)

- Designed and implemented Synchronous and Asynchronous FIFO in Verilog HDL, supporting configurable depth and width with full/empty flag generation, write/read pointer logic, and overflow/underflow protection for reliable data transfer.
- Verified functionality and robustness through simulations and testbenches.

LANGUAGES

English, Hindi & Marathi

ADDITIONAL INFORMATION

Awarded the prestigious Nurturing Brilliance Cummins Scholarship (2018-2021) by Cummins India, recognizing meritorious students across engineering disciplines.